

Parametrizing solutions of $R\vec{\xi} = 0$

not parametrizable



$$R\vec{\xi} = 0$$

add ghost task(s)



constrain

$$\xi_{GT} = 0$$

minimally parametrized (Ghost tasking)

$$(R \quad -\Lambda) \begin{pmatrix} \vec{\xi} \\ \xi_{GT} \end{pmatrix} = 0$$

constrain

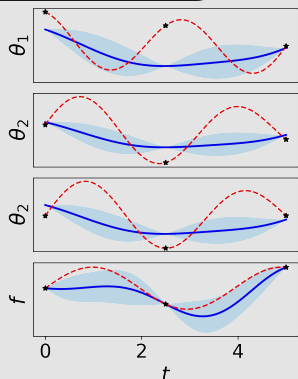
$$\xi_{GT} = 0$$

$$R\vec{\xi} = 0 \Leftrightarrow \mathcal{L}\vec{u} = \vec{f}$$

over parametrized

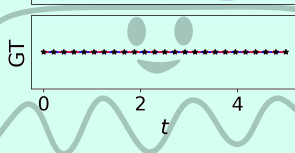
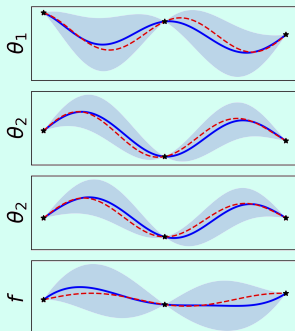
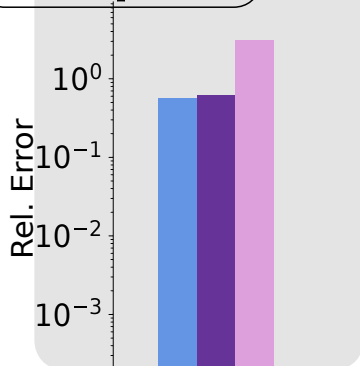
$$(\mathcal{L} \quad -\mathbb{I}_q) \begin{pmatrix} \vec{\xi} \\ f_1 \\ \vdots \\ f_q \end{pmatrix} = 0$$

Forward problem



— Mean
 ±2σ confidence
 - - Truth
 ★ Training data

Inverse problem



Parameter 1
 Parameter 2
 Ratio

Parameter 1
 Parameter 2
 Ratio

